

# Analysis of Architectural Refinements and Mathematical Optimization for Hybrid Mamba-xLSTM Medical Foundation Models

The transition of medical artificial intelligence from large-scale, general-purpose models toward specialized, efficient architectures requires a rigorous re-evaluation of the foundational components governing cross-modal alignment and sequence modeling. The current analysis focuses on a 70M parameter hybrid model that integrates the selective state-space mechanisms of Mamba with the matrix-valued memory structures of Extended Long Short-Term Memory (xLSTM).<sup>1</sup> This specific hybrid configuration, often termed "Mamba-xLSTM" or "xMamba," is designed to exploit the linear scaling properties of State-Space Models (SSMs) while maintaining the robust gating and high-capacity memory associated with the latest iterations of LSTM technology.<sup>3</sup> Initial stage-0 evaluations of this architecture, conducted on biomedical corpora such as PubMed, reveal a highly efficient base for language modeling with weights totaling approximately 83.1 million parameters.<sup>1</sup> Performance metrics on validation sets show a perplexity (PPL) of 13.10 and a cross-entropy loss of 2.5729, with significant computational throughput reaching 88,076 tokens per second on NVIDIA A100-SXM4-40GB hardware.<sup>1</sup> Despite these promising metrics for unimodal processing, the move toward stage-1 multimodal alignment—aligning medical imagery (e.g., chest X-rays) with radiology reports—uncovered five distinct architectural and mathematical gaps that must be resolved to ensure clinical factuality and representational robustness.<sup>3</sup>

## The Hybrid Mamba-xLSTM Backbone and Stage-0 Baseline

The baseline 70M model utilizes a layer pattern specified as ['mamba', 'mamba', 'mlstm', 'mamba', 'mamba', 'mlstm', 'mamba', 'mamba'], featuring a hidden dimension of 512 across eight layers.<sup>1</sup> The inclusion of matrix-valued LSTM (mLSTM) layers is a critical departure from purely additive architectures, as it allows the model to store information in a matrix format, facilitating more complex interaction between temporal states than traditional vector-based hidden states.<sup>1</sup> The Mamba layers provide a selective scan mechanism (S6) that filters irrelevant information from long-range clinical sequences, while the mLSTM layers introduce an exponential gating mechanism and a covariance-like memory update rule.<sup>1</sup>

## Evaluation of Stage-0 Language Modeling Performance

The throughput measurements across varying sequence lengths demonstrate the hardware

efficiency of the hybrid architecture. At shorter sequence lengths, the model maintains high utilization, which remains consistent as the sequence length doubles, indicating the linear scaling benefits of the Mamba component.<sup>1</sup>

| Sequence Length | Throughput (Tokens/s) | Batch Latency (ms) |
|-----------------|-----------------------|--------------------|
| 128             | 82,541                | 24.8               |
| 256             | 86,984                | 47.1               |
| 512             | 87,253                | 93.9               |
| 1024            | 88,076                | 186.0              |

The peak GPU memory utilization of 7.67 GB during the evaluation phase suggests that this 70M model is well-suited for edge-deployment in clinical settings where VRAM might be limited compared to high-performance computing clusters.<sup>1</sup> However, the current stage-0 model serves primarily as a language backbone. The challenge arises when this backbone is integrated into a Vision-Language Processing (VLP) framework, where it must align its latent space with visual features extracted from medical imaging encoders.<sup>3</sup>

## Gap 1: Addressing Low Negative Coverage in Contrastive Learning

Contrastive learning for medical vision-language alignment typically utilizes the NT-Xent (Normalized Temperature-scaled Cross Entropy) loss, a form of the InfoNCE objective.<sup>6</sup> For an encoded query  $q$  and a set of dictionary keys  $k$ , the goal is to align the query with its corresponding positive key  $k^+$  while maximizing its distance from a set of  $K$  negative keys  $\{k_i\}$ .<sup>8</sup>

### Mathematical Formulation of the Coverage Deficit

The loss for a specific query  $q$  is defined as:

$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k^+ / \tau)}{\sum_{i=0}^K \exp(q \cdot k_i / \tau)}$$

In this formulation, the denominator is a summation over one positive and  $K$  negative samples.<sup>7</sup> In standard batch-based contrastive learning (e.g., SimCLR),  $K$  is restricted to  $N - 1$ , where  $N$  is the batch size.<sup>9</sup> For the 70M medical model, batch sizes are often constrained by the memory-intensive nature of 3D medical imaging or high-resolution radiology scans, even when the Mamba-xLSTM backbone is itself efficient.<sup>11</sup> When  $N$  is small, the model suffers from low negative coverage, meaning it does not encounter a diverse enough set of "hard negatives" to learn the fine-grained discriminative features necessary for medical diagnosis.<sup>14</sup>

The gradient of the InfoNCE loss with respect to the query  $q$  reveals the impact of this limitation:

$$\frac{\partial \mathcal{L}_q}{\partial q} = \frac{1}{\tau} \left( \sum_{i=0}^K P_i k_i - k^+ \right)$$

where  $P_i = \frac{\exp(q \cdot k_i / \tau)}{\sum_{j=0}^K \exp(q \cdot k_j / \tau)}$ .<sup>16</sup> With a small  $K$ , the probability distribution  $P$  is poorly estimated, and the "push" provided by the negative samples  $\sum P_i k_i$  is insufficient to prevent representational collapse or to distinguish between highly similar clinical conditions, such as different types of interstitial lung disease.<sup>15</sup>

## Architectural Fix: Momentum Contrast (MoCo) and Dynamic Queues

To solve the negative coverage gap without increasing the hardware-limited batch size, the architecture should be modified to incorporate a dynamic queue and a momentum-based encoder update mechanism, as proposed in the MoCo framework.<sup>8</sup> This decouples the dictionary size  $K$  from the mini-batch size  $N$ .<sup>8</sup>

1. **Queue Mechanism:** A FIFO (First-In, First-Out) queue is maintained to store the feature embeddings from previous mini-batches. This allows the model to utilize  $K = 65,536$  or more negative samples even with a batch size of  $N = 256$ .<sup>8</sup>
2. **Momentum Encoder:** To maintain consistency of the embeddings in the queue as the model parameters  $\theta_q$  evolve, a separate key encoder  $\theta_k$  is used. This encoder is

updated via an exponential moving average:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

where  $m \in [0.99, 0.9999]$  is a momentum coefficient.<sup>8</sup>

| Feature                | Standard SimCLR                   | MoCo (Fix)                     |
|------------------------|-----------------------------------|--------------------------------|
| Negative Sample Pool   | Limited to Mini-batch ( $N - 1$ ) | Large Dynamic Queue ( $K$ )    |
| Dictionary Consistency | Updates every step                | Momentum-stabilized            |
| Hardware Dependency    | Requires large GPUs for large $N$ | Memory-efficient for small $N$ |
| Negative Mining        | Stochastic within batch           | Sustained coverage over epochs |

This architectural fix ensures that the 70M hybrid model can learn from a rich distribution of clinical negatives, improving the model's ability to distinguish subtle pathological features in chest X-rays that might otherwise be blurred into a single "abnormal" cluster.<sup>3</sup>

## Gap 2: Space Mismatch and the Collapse of Identity Projections

A significant error in many cross-modal models is the assumption that if the dimensions of the image encoder and text encoder are matched (e.g., both at  $d = 512$ ), they can be aligned directly using an identity projection.<sup>18</sup> In the context of the 70M hybrid model, an identity projection  $g(x) = x$  forces the backbone Mamba-xLSTM layers to directly minimize the contrastive loss, which often leads to "dimensional collapse" or "backbone corruption".<sup>20</sup>

## Theoretical Basis for Non-linear Projection Heads

Identity projections are mathematically constrained by the Lipschitz constant of the final layers and the spectral properties of the backbone.<sup>22</sup> If the backbone is forced to satisfy the contrastive requirement  $\text{sim}(q, k) \approx 1$  for positive pairs using a linear or identity map, it may discard general semantic information to prioritize the high-frequency features that minimize the contrastive loss most efficiently.<sup>20</sup>

An analysis of the implicit bias in training algorithms reveals that features become increasingly unequal in deeper layers.<sup>20</sup> Lower layers tend to maintain normalized and less specialized representations, while the final layer under an identity projection becomes overly specialized to the pre-training objective, losing its generalizability for downstream tasks like zero-shot classification or report generation.<sup>20</sup>

## Mathematical Fix: Non-linear MLP Projector

The solution is to insert a shallow, non-linear MLP (Multi-Layer Perceptron) projection head  $g(\cdot)$  between the backbone and the contrastive loss.<sup>10</sup> The projection head is typically a 2-layer MLP with a hidden dimension larger than the input (e.g., 2048 for a 512-dim input) and a ReLU or GeLU activation.<sup>10</sup>

The projection is formulated as:

$$z = W_2 \sigma(W_1 h)$$

where  $h$  is the pooled representation from the Mamba-xLSTM backbone,  $\sigma$  is the non-linear activation, and  $z$  is the vector used for the contrastive loss.<sup>20</sup>

| Component                 | Identity Map                          | MLP Projection Head (Fix)                          |
|---------------------------|---------------------------------------|----------------------------------------------------|
| Representational Capacity | Strictly Linear                       | Non-linear latent warping                          |
| Backbone Protection       | None (Backbone is directly optimized) | Buffer (Head absorbs task-specific specialization) |
| Feature Diversity         | Low (Risk of collapse)                | High (Encourages diverse                           |

|                     |            |                                               |
|---------------------|------------|-----------------------------------------------|
|                     |            | feature learning)                             |
| Downstream Transfer | Suboptimal | Superior (using pre-projection features $h$ ) |

By using an MLP head during pre-training and discarding it for downstream tasks, the 70M model preserves its internal Mamba-xLSTM feature diversity while allowing the MLP to handle the specific geometric transformations required for cross-modal alignment.<sup>20</sup>

## Gap 3: Transitioning from Mean Pooling to Attention-Weighted Pooling

In the baseline configuration, the model often employs mean pooling to aggregate the sequence of hidden states  $h_t$  into a single vector  $h_{global} = \frac{1}{T} \sum_{t=1}^T h_t$ . However, clinical text, especially radiology reports, is characterized by highly localized information and the frequent use of negation.<sup>2</sup> Mean pooling treats all tokens—including uninformative filler words and complex negated findings—with equal weight, which leads to "semantic dilution".<sup>3</sup>

### The BioViL-T Approach to Spatiotemporal Aggregation

BioViL-T, a domain-specific vision-language model for radiology, emphasizes the importance of capturing temporal structure and semantic nuances.<sup>3</sup> For a hybrid model processing medical data, mean pooling fails to account for the fact that a single sentence like "no evidence of pneumothorax" is fundamentally more important than several sentences of boilerplate text.<sup>2</sup>

To address this, the 70M model should implement attention-weighted pooling (also referred to as "attention pooling" or "transformer pooling").<sup>11</sup> This mechanism learns to assign importance weights  $\alpha_t$  to each state in the sequence:

1. **Query Definition:** A learnable query vector  $q_{pool}$  is introduced.
2. **Weight Computation:**

$$\alpha_t = \text{softmax} \left( \frac{(q_{pool} W_Q)(h_t W_K)^T}{\sqrt{d}} \right)$$

3. **Aggregation:**

$$h_{attn} = \sum_{t=1}^T \alpha_t(h_t W_V)$$

where  $W_Q, W_K, W_V$  are learnable projection matrices.<sup>27</sup>

### Impact on Clinical Negation and Phrase Grounding

| Pooling Method    | Response to "No evidence of pneumonia"                           | Temporal Progression Handling                        |
|-------------------|------------------------------------------------------------------|------------------------------------------------------|
| Mean Pooling      | Averages "no" and "evidence" with all other report tokens.       | Treats prior and current images as an average blend. |
| Attention Pooling | Can attend heavily to "no" and "pneumonia" as a unified finding. | Learns to weight changes in findings across time.    |

This architectural fix allows the 70M model to perform "phrase grounding"—associating specific words in a report with localized regions in an image—far more effectively than mean pooling.<sup>3</sup> It also handles missing images or prior reports by learning to assign near-zero weights to padding tokens or uninformative states.<sup>2</sup>

## Gap 4: Correcting Dropout Mismatch via R-Drop Regularization

Dropout is a ubiquitous regularization technique used to prevent overfitting in deep models like the hybrid Mamba-xLSTM.<sup>1</sup> However, dropout introduces a stochastic discrepancy: during training, the model acts as a "sub-model" with randomly deactivated neurons, whereas at inference, the "full model" is used.<sup>31</sup> This leads to an inconsistency in the output distributions, which can be particularly damaging in medical reasoning where high-precision classification is required.<sup>30</sup>

### The Mathematical Inconsistency of Dropout

Let  $f(x, \xi)$  be the model output for input  $x$  given a random dropout mask  $\xi$ . During training, we minimize  $\mathcal{L}(f(x, \xi), y)$ . During inference, we use the expected output

$E_{\xi}[f(x, \xi)] \approx f(x, E[\xi])$ , which is the full model.<sup>32</sup> The variance between  $f(x, \xi_1)$  and  $f(x, \xi_2)$  for different masks  $\xi$  represents a source of training instability.<sup>32</sup>

## Solution: R-Drop (Regularized Dropout)

To solve the train-eval mismatch, the 70M model should implement R-Drop.<sup>31</sup> In each training step, every data sample  $x_i$  is passed through the model twice, with two different random dropout masks  $\xi_1$  and  $\xi_2$ .<sup>32</sup> This yields two output distributions,  $P_1(y|x_i)$  and  $P_2(y|x_i)$ . The R-Drop objective adds a symmetric Kullback-Leibler (KL) divergence term to the loss:

$$\mathcal{L}_{KL}^i = \frac{1}{2} (D_{KL}(P_1 \| P_2) + D_{KL}(P_2 \| P_1))$$

The total loss for the mini-batch becomes:

$$\mathcal{L}_{total} = \frac{1}{N} \sum_{i=1}^N (\mathcal{L}_{NLL}(P_1, y_i) + \mathcal{L}_{NLL}(P_2, y_i) + \alpha \cdot \mathcal{L}_{KL}^i)$$

where  $\alpha$  is a weight coefficient.<sup>31</sup>

| Feature              | Standard Dropout            | R-Drop (Fix)                             |
|----------------------|-----------------------------|------------------------------------------|
| Training Objective   | $\mathcal{L}_{NLL}$ only    | $\mathcal{L}_{NLL}$ + Output Consistency |
| Forward Passes       | 1 per sample                | 2 per sample                             |
| Model Generalization | Implicit Ensemble           | Explicit Consistency between sub-models  |
| Medical Context      | Prone to distribution shift | Robust to noise and imputation           |

In medical VLP, where a single input might be "augmented" through dropout to act as its own positive pair (as in SimCSE), R-Drop ensures that the learned representation is invariant to the



specific neurons dropped, making the 70M model significantly more robust to the inherent noise in clinical data.<sup>30</sup>

## Gap 5: Thermal Stability and Logit Scale Clipping

The final gap concerns the learnable temperature parameter  $\tau$  used in the contrastive loss. In models like CLIP, the similarity between image embedding  $v$  and text embedding  $t$  is computed as  $s = \frac{v \cdot t}{\tau}$ .<sup>34</sup> Many frameworks treat  $\tau$  as a learnable parameter, typically by optimizing  $\log(1/\tau)$  (the logit scale).<sup>34</sup>

### Instability of Unconstrained Learnable Temperature

In an unconstrained setting, the optimization of the cross-entropy loss naturally drives the temperature  $\tau$  toward zero to maximize the "sharpness" of the probability distribution for correct pairs.<sup>7</sup> As  $\tau \rightarrow 0$ , the similarity scores are scaled by an increasingly large factor, which can lead to numerical overflow, gradient explosion, and extreme over-penalization of negative samples.<sup>16</sup> This is particularly problematic in medical VLP where the dataset may contain mislabeled or noisy pairs; a very low temperature forces the model to overfit to these noisy samples.<sup>34</sup>

### Fix: Logit Scale Clipping and Initialization

To ensure training stability in the 70M model, the learnable logit scale must be clipped.<sup>35</sup> The CLIP paper found that clipping the logit scale at 100 (corresponding to a minimum  $\tau = 0.01$ ) was necessary to prevent instability.<sup>35</sup>

#### Implementation Details:

- Initialization:** Initialize the logit scale to  $\log(1/0.07) \approx 2.6592$ , as per the original CLIP implementation.<sup>26</sup>
- Clipping Rule:**

$$\text{logit\_scale} = \text{clamp}(\text{logit\_scale}, \text{max} = \log(100))$$

This operation is applied after the optimizer step in each training iteration.<sup>35</sup>

| Parameter | Initial Value | Maximum Bound | Purpose |
|-----------|---------------|---------------|---------|
|           |               |               |         |

|                        |        |                 |                                       |
|------------------------|--------|-----------------|---------------------------------------|
| $\tau$                 | 0.07   | N/A (Min: 0.01) | Stabilizes probability distribution   |
| $1/\tau$ (Logit Scale) | 14.28  | 100             | Prevents gradient explosion           |
| $\log(1/\tau)$         | 2.6592 | 4.6052          | Ensures stable parameter optimization |

This constraint acts as a "temperature floor," preventing the model from becoming excessively confident in potentially noisy medical pairings and ensuring that the gradient signal remains informative throughout the training process.<sup>7</sup>

## Synthesis of Integrated Fixes for the 70M Hybrid Model

The identified gaps and their respective fixes are not independent; rather, they form an integrated framework for robust medical representation learning. The high throughput of the stage-0 Mamba-xLSTM model (88k tokens/s) provides the computational headroom required for more complex pooling and consistency regularization like R-Drop, which requires two forward passes.<sup>1</sup>

### Summary of Architectural Improvements

| Component       | Baseline Status       | Proposed Architectural Fix | Mathematical Justification                           |
|-----------------|-----------------------|----------------------------|------------------------------------------------------|
| Negative Pool   | Small (Batch-limited) | MoCo Dynamic Queue         | Minimizes gradient variance in InfoNCE. <sup>8</sup> |
| Projection Head | Identity/Linear       | 2-Layer Non-linear MLP     | Prevents dimensional collapse and backbone           |

|                 |                  |                       |                                                                 |
|-----------------|------------------|-----------------------|-----------------------------------------------------------------|
|                 |                  |                       | corruption. <sup>20</sup>                                       |
| Feature Pooling | Global Mean      | Attention-Weighted    | Captures localized clinical nuance and negation. <sup>3</sup>   |
| Regularization  | Standard Dropout | R-Drop (Symmetric KL) | Alleviates training-inference distribution shift. <sup>31</sup> |
| Logit Scaling   | Unconstrained    | Logit Scale<br>$\leq$ | Ensures numerical stability and noise tolerance. <sup>35</sup>  |

The interaction between Gap 1 (Negative Coverage) and Gap 5 (Temperature) is particularly noteworthy. A large pool of negatives in a MoCo queue allows the model to utilize a higher temperature effectively, as there are more samples to "push" against, reducing the risk of overfitting.<sup>15</sup> Conversely, the attention-weighted pooling (Gap 3) provides higher-quality positive representations, which allows the MLP head (Gap 2) to map modalities into a shared space where the contrastive loss is more semantically grounded.<sup>3</sup>

### Future Outlook and Implementation Considerations

For the 70M parameter scale, these fixes are essential because smaller models have less "emergent" capacity to overcome architectural flaws than larger foundation models.<sup>13</sup> The computational overhead of R-Drop (two forward passes) and attention pooling is offset by the linear efficiency of the Mamba and xLSTM layers.<sup>1</sup> Implementing the MoCo queue further optimizes resource usage by allowing high-quality training on a single GPU (e.g., A100 40GB) that would otherwise require multi-GPU clusters for equivalent large-batch SimCLR training.<sup>1</sup>

By systematically applying these refinements, the 70M Hybrid Mamba-xLSTM model can move from a simple language baseline to a robust medical reasoning engine, capable of accurate zero-shot inference, clinical phrase grounding, and factually consistent radiology report generation.<sup>3</sup> The rigorous mathematical grounding of these solutions ensures that the model remains stable across diverse medical datasets and varying clinical conditions, paving the way for more reliable and efficient medical AI systems.

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